

Chapter 7

Controlling Access to Files

Effects of Permissions on Files and Directories

Permission	Effect on files	Effect on directories
r (read)	File contents can be read.	Contents of the directory (the file names) can be listed.
w (write)	File contents can be changed.	Any file in the directory can be created or deleted.
x (execute)	Files can be executed as commands.	The directory can become the current working directory. (You can cd into it, but also require read permission to list files found there.)

- is a regular file.

d is a directory.

l is a soft link.

```
[root@localhost ~]# ll
total 0
-rw-r--r--. 1 root root 0 Sep  4 18:09 File
drwxr-xr-x. 2 root root 6 Sep  4 18:09 Folder
```

Managing File System Permissions from the Command Line

Changing Permissions with the Symbolic Method

- Who is **u**, **g**, **o**, **a** (for **user**, **group**, **other**, **all**)
- What is **+**, **-**, **=** (for **add**, **remove**, **set exactly**)
- Which is **r**, **w**, **x** (for **read**, **write**, **execute**)

```
[user@host ~]$ chmod go-rw file1
```

```
[user@host ~]$ chmod a+x file2
```

```
[root@localhost ~]# ll
total 0
-rw-r--r--. 1 root root 0 Sep  4 18:09 File
drwxr-xr-x. 2 root root 6 Sep  4 18:09 Folder
[root@localhost ~]# chmod g+rw- File
[root@localhost ~]# ll
total 0
-rw-rw-r--. 1 root root 0 Sep  4 18:09 File
drwxr-xr-x. 2 root root 6 Sep  4 18:09 Folder
[root@localhost ~]#
```

Changing Permissions with the Numeric Method

In the example below the # character represents a digit.

```
chmod ### file|directory
```

```
[user@host ~]$ chmod 644 samplefile
```

```
[user@host ~]$ chmod 750 sampledirt
```

Changing File and Directory User or Group Ownership

```
[root@host ~]# chown student test_file
```

```
[root@host ~]# chown visitor:guests test_dir
```

```
[root@host ~]# chown owner.group filename
```

Default permissions

The default permission for a new file or directory is determined by applying the *umask* to the base permission

The default *umask* for a root user is **0022**

```
[root@localhost ~]# umask
0022
[root@localhost ~]#
```

To change Umask

```
[root@localhost ~]# umask 0002
[root@localhost ~]# umask
0002
[root@localhost ~]#
```

To Change umask Permanently

```
[root@localhost ~]# vi .bashrc
```

```
# User specific environment
if ! [[ "$PATH" =~ "$HOME/.local/bin:$HOME/bin:" ]]
then
    PATH="$HOME/.local/bin:$HOME/bin:$PATH"
fi
export PATH

# Uncomment the following line if you don't like systemctl's auto-paging feature:
# export SYSTEMD_PAGER=

# User specific aliases and functions

alias rm='rm -i'
alias cp='cp -i'
alias mv='mv -i'
umask 0002
```